

Has (multi-)factor investing become obsolete?

Tim Herpich*
(Dated: October 31, 2020)

I. INTRODUCTION

The term *factor premia* refers to characteristics of assets, which can explain the risk-return relationship within an asset class. The so-called *risk factors* have been statistically identified as the driving forces underlying these premia. While the duality of risk premia and risk factors is believed to be present in all major asset classes, we will in the following focus on stocks, for which the body of academic literature is arguably the largest regarding this topic. Historically, the discovery and identification of a multitude of risk factors helped resolving the apparent anomaly of different returns observed among well-diversified stock portfolios.

At the beginning of the second half of last century, the *Capital Asset Pricing Model* (CAPM) was the primary asset pricing model. CAPM makes predictions about the expected excess return of stocks relative to the risk-free investment using only the stock's volatility in relation to the overall stock market, coined as β . Empirical studies of historical stock returns, however, showed that the explanatory power of CAPM is only about 65%, *i.e.* there could be significant deviations observed in well-diversified portfolios. The analysis of historical stock market data furthermore revealed that the small-size (small free-float-adjusted market capitalization compared to market average) stocks [1] and value (low price-to-earnings/bookvalue figures compared to business sector average) stocks [2] had consistently higher average returns as predicted by the market β . These observations advocated the existence of α , *i.e.* excess risk-adjusted returns, and thus apparently falsified the CAPM and its underlying dogma of efficient markets.

Fama and French reconciled these apparent contradictions by extending the CAPM, a one-risk-factor model, towards a three-factor model which also included the risk factors that had been identified as driving forces behind the *Small-Size* and *Value premia* [3]. The explanatory power of the three-factor model has been empirially shown to be about 90%. In the groundbreaking paper [4], Carhart further generalized the three-factor model to a four-factor one including the *Momentum* risk factor and Marx added the *Quality* factor [5] to formulate a five factor model exhibiting an explanatory power of well above 95%. The Momentum factor is to be taken literally and refers to stocks that displayed a high price momentum in the past (typically for a period of up to one year). Moreover, here, the term Quality refers to

the balance sheet of the public company, *i.e.* it is associated with stocks that display more stable earnings and higher profit margins than the immediate competitors in their business sector. As a side remark, Fama and French proposed an alternative five-factor model excluding the momentum risk factor [6].

The recent underperformance of the (global) value stocks compared to the overall market has received a lot of media attention: Since 2005 the MSCI World IMI Value has been trailing the MSCI World IMI - a proxy index for the global investable stock market, which covers $\sim 99\%$ of the global free-float-adjusted market capitalization - by 1.4% per year on average. If considering only the past three years (2017–2020), the underperformance has even been 5.4% per year on average. This observation prompted fierce debates on the continued existence of value, and more generally factor, premia post their publication among (professional) investors and (self-declared) stock market “experts”. This work is devoted to investigating, whether it can scientifically be concluded that the value premium has vanished post-publication. To this end, we will study the available historical data of risk premia in details, address pertinent peer-reviewed literature and draw a conclusion.

II. PRELIMINARIES

First, we want to emphasize that 300 risk factors have been identified in the body of academic literature that is estimated to encompass a high four-figure amount of papers. Yet, most of these risk factors were shown to be merely a combinatoric product of the elementary ones. In our opinion, academia's consensus is that the above-mentioned factors (Value, Size, Momentum, Quality) and *Political Risk* constitute the fundamental core of the risk-factor universe [7]. For completeness, we add that the Political Risk factor is driving the returns of companies' stocks that operate in countries which are not fully developed capitalistic and democratic systems. In this work, we will only consider the arguably most famous factors: Equity, Small-Size and Value, while the focus will be on the Value factor to address its critics.

To fix terminology, we will not use the popular expression of smart beta factors, which refers to the risk factors other than the traditional equity risk factor β . To fix methodology, when explicitly stating figures related to risk premia, we will implicitly refer to the *Long-only* risk premia. The reason is that investable funds that are compatible with the consolidated EU directive *UCITS* and thus accessible for European investors can legally only implement the long position in stocks exhibiting the de-

* Electronic Mail: tim.herpich@d-fine.de

sired, risk-factor related, characteristics but not the short position in stocks displaying the opposite features to that characteristics. The Long-only risk premia are naturally smaller than the Long-short ones, but are more meaningful in practice for the above-stated reason.

III. DATA ANALYSIS

1. Negative factor premia as the “historic norm”

It is natural to ask the question, if the recent underperformance of the value-factor strategy compared to the overall market return is a historically singular event. For this purpose, let us in the following consider the US stock market, as its the historical data including factor premia is available for a much larger period than for global, non-US markets. First, while the equity premium β , that is the excess return of the market with respect to a risk-free investment, has been 6.28% per year on average from 1926 to 2018, the small-size and value risk premia, that is the excess return of small and values stocks relative to the overall US stock market, have been 1.88% and 2.9% per year on average, respectively, over the same period of time [8].

	Equity	Small-size	Value
1 year	30%	43%	39%
5 year	21%	37%	25%
10 year	15%	27%	16%

Table I. Percentage of time periods with negative factor premia for the US stock market from 1926 until 2018. Data provided by [9]. The figures are before costs of investment and taxes.

However, as shown in table I illustrating the equity-, small-size- and value premia for different time periods in the US stock market for approximately the past 90 years, there have been repeatedly (extended) periods of negative factor premia during 1926–2018: The equity premium has been negative in 15% of the 10-year periods, that is the US stock market delivered a lower return than a risk-free investment over 10 years. Furthermore, the small-size and value premia have been negative in 27% and 16% of the considered 10-year periods. This clearly shows that negative factor premia have occurred multiple times throughout the past decades and the recent negativity of the value premium is therefore by no means a singularity but rather to be expected at times.

There are a few additional remarks to be made, here. First, negative small-size- or value premia do not necessarily imply a negative return, but only an underperformance with respect to the return of the overall stock market. Secondly, in view of diversification, it is interesting to observe that in many cases of periods of under-average equity returns the value premium has been positive during the same period (not shown in table I). Thirdly, it is noteworthy that the number of periods with negative

risk premia is decreasing as the time period under consideration increases.

2. Regression to the mean

It can be empirically observed that the stock market displays a so-called *regression to the mean*, which refers to the phenomenon that, compared to the historical average, periods of small equity premia tend to be preceding periods with equity premia that are higher than the historical average and vice versa. In other words, the expected return of the stock market in the future is a function of the current fundamentals underlying the stock market. The arising question is, if that this principle also applies to the other factor premia. Indeed, Fig. 1 depicting the annualized factor premia for a ten-year period following a ten-year period of negative factor premia, suggests that small- and value premia tend to strongly rebound into the positive regime after a period of negative returns. Strikingly, the worst ten-year-period performance of the value factor was still 2.9% above the annual return of the overall stock market. We conclude from the figure that investors who continued with their risk-factor strategy were often generously rewarded after a period of negative factor premia.

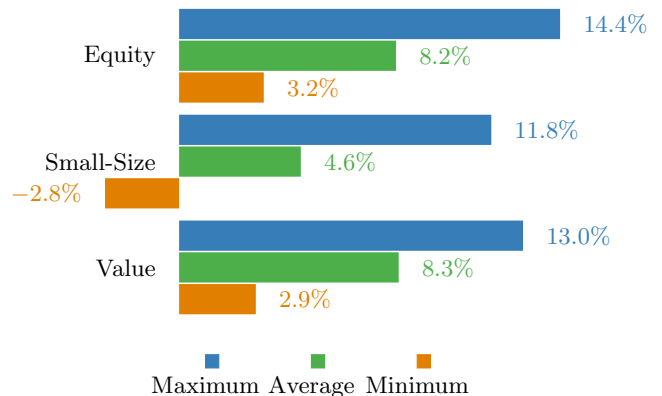


Figure 1. Annualized factor premia over a ten-year period given that the previous ten-year period yielded negative factor premia for the US stock market from 1926 until 2018. Data taken from [9]. The figures are before costs of investment and taxes.

3. Peak of Value Spread

Another argument along those lines, which is in favor of the continuation of the value premium, is the sharp increase of the value spread, that is the ratio of the price-to-book value for value and growth stocks, over the past two years [10]. This is a useful metric as it indicates, if the performance of value stocks relative to

growth ones is due to their price movement only or due to the underlying fundamentals. In fact, Fig. 2 illustrates that, for the given methodology used to define the value spread (see Appendix in Ref. [10]), the spread has only been three time higher in the past than today: in the early 90's, during "New-Economy Crisis" (also "Dot-Com Bubble") around the millenium change and at the peak of the "Subprime-Mortgage Crisis".

A closer inspection shows that, statistically, the value spread is currently the 96th (99th if you exclude the

Dot-Com Bubble) percentile versus history. If alternative methods to determine the value spread are chosen, the value stocks appear to be even cheaper, as the value spread can be the highest throughout the considered time period, apart from the Dot-Com Bubble. Just after the latter, value stocks exhibited a period of exceptionally strong returns. Thus, the value spread indicates that value stocks are currently valued at historically inexpensive levels and, empirically, have a considerable probability to deliver higher returns than the market in the future.

Academic HML (B/P) Spread
December 1964 – August 2019

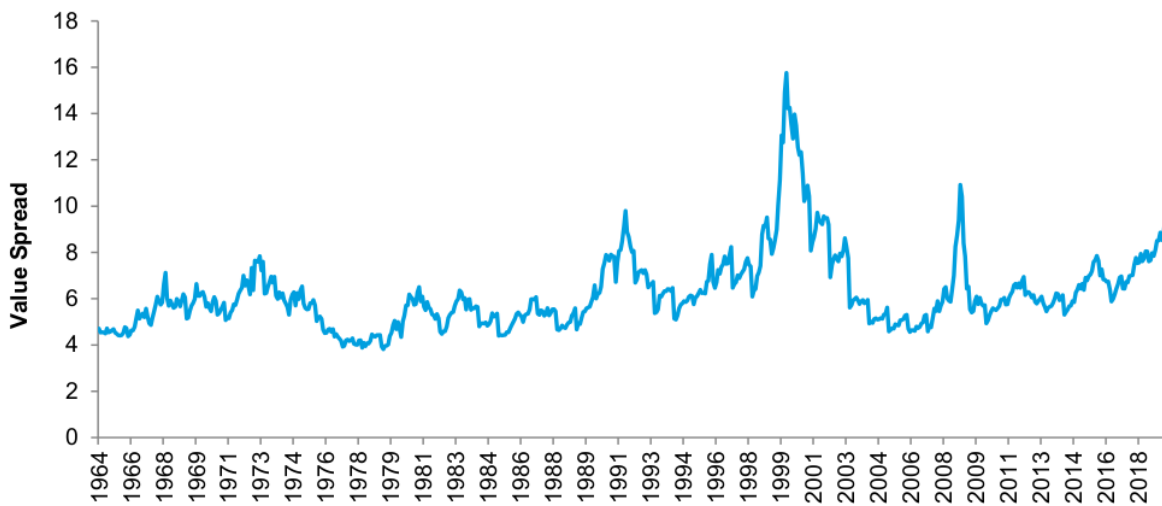


Figure 2. Figure taken from Ref. [10]. From 1964-2019, Price-to-Book values of the expensive 30% of stocks divided by the Price-to-Book values of the inexpensive 30% of stocks are depicted. This corresponds to the so-called "academic" method to determine the value spread. For further details regarding the methodology see Appendix in Ref. [10].

4. Sampling Methods

Not surprisingly, Fama and French also addressed the question, whether the small and value premia vanished post their publication [3]. In Ref. [11], they applied the Bootstrap method on the US stock market data from 1963-2016 to simulate 10^6 time periods of varying length in order to model the underlying distribution of stock returns and therefore gain access to the distribution of factor premia. Remarkably, they obtained the result that the probability for a positive combined small-value premium (cf. Sec. IV) is *larger* than the probability for a positive equity premium. In other words, using standard sampling techniques, it appears more likely that a diversified portfolio with a tilt towards small and value US

companies is outperforming the US stock market than the US stock market beating a risk-free investment.

IV. CONCLUSION

To sum up, it is evident from the historical data that there have always been periods spanning over more than ten years, where the risk premia have been negative. In that regard, any general statements, including the call for the end of (value-)factor investing, based on up-to-ten-year periods can not be conclusive. To the contrary, we believe that the unpredictability (on short time scales) of the evolution of factor premia with respect to a reference premium, for instance equity premium or risk-free

investment premium, represents the *risk* associated with that factor premium and (extended) periods of underperformance corresponds to a *manifestation* of exactly that risk. In the long run, the additional risk accepted *and endured* throughout very volatile times by the investor has to be compensated in form of higher premia: That is the fundamental argument for stock returns being higher than the return of a risk-free investment. The same should apply to the returns of value and small-cap stocks since they have the same underlying theory and model, which have been empirically confirmed by the available data as shown above.

Returning to the five-factor model from above, we note that the Momentum and Quality factor have significantly outperformed the overall stock market over both the past five and ten years, that is during the same time where small-cap and value factors have been trailing the stock market. Therefore, and to mitigate the additional risks due to factor investing, it seems advisable to diversify over several risk factors. In theory, this can be done in two ways: First, via a Linear Multifactor (also Fund-Level Multifactor) portfolio which replicates multiple single-factor indices indepently. Secondly, via an In-

tegrated Multifactor (also Stock-Level Multifactor) portfolio which replicates a single multi-factor index. Recent studies suggest that the latter approach is superior [12].

In this work, we addressed the question about the continued existence of factor premia other than the equity one using the example of the value factor. As specified in Sec. II, we considered Long-Only premia as they are practically meaningful for European investors who have only access to equity funds complying with the *UCITS* directive. Yet, there is still a gap to be bridged between theoretical models that motivate the creation of (multi-)factor stock indices and their transformation into accessible/investable products, such as *Exchange-Traded-Funds* (ETFs). In this regard, we are critical as there are currently only few accessible European ETFs that track single-factor indices and only two that aim at replicating a multi-factor index. Apart from the currently little competition among ETF firms in this field, we are concerned about whether these few ETFs are properly implemented for various reasons that shall not be further elaborated here. According to both experts and the firms themselves, there are more products and competition to be expected in the future.

-
- [1] Rolf W. Banz, “The relationship between return and market value of common stocks,” *Journal of Financial Economics* **9**, 3 – 18 (1981).
 - [2] Barr Rosenberg, Kenneth Reid, and Ronald Lanstein, “Persuasive evidence of market inefficiency,” *The Journal of Portfolio Management* **11**, 9–16 (1985).
 - [3] E. F. Fama and K. R. French, “The cross-section of expected stock returns,” *The Journal of Finance* **47**, 427–465 (1992).
 - [4] Mark M. Carhart, “On persistence in mutual fund performance,” *The Journal of Finance* **52**, 57–82 (1997).
 - [5] Robert Novy-Marx, “The other side of value: The gross profitability premium,” *Journal of Financial Economics* **108**, 1 – 28 (2013).
 - [6] Eugene F. Fama and Kenneth R. French, “A five-factor asset pricing model,” *Journal of Financial Economics* **116**, 1 – 22 (2015).
 - [7] We intentionally omit the factor *Low/Minimum Volatility* which is also considered as an independent, elementary factor. The reason is that its premium technically (not economically!) relies on the exclusion of small growth stocks, *i.e.* on a strategy that can also be implemented via a small-value multifactor index (cf. Sec. IV).
 - [8] Elroy Dimson, Paul Marsh, and Mike Staunton, *Credit Suisse Global Investment Returns Yearbook 2018* (Credit Suisse Research Institute, 2019) p. 256 pages.
 - [9] “Dimensional fund advisors, fama & french, ibbotson associates, s&p, morningstar direct, crsp und compustat.”.
 - [10] C. Asness, “It’s time for a venial value-timing sin,” <https://www.aqr.com/Insights/Perspectives/Its-Time-for-a-Venial-Value-Timing-Sin>, accessed: 2020-9-30.
 - [11] E. F. Fama and K. R. French, “Volatility lessons,” *Chicago Booth Research Paper No. 17-33* (2018), <https://doi.org/10.2139/ssrn.3081101>.
 - [12] S. Fitzgibbons, J. Friedman, L. Pomorski, and L. Serban, “Long-only style investing: Don’t just mix, integrate,” *The Journal of Investing* **26** (2017), <https://dx.doi.org/10.2139/ssrn.2802849>.